Ecuaciones Principales

$$F_a(t) = F_2(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_2(t) = \frac{P_a(t) - P_p(t)}{2}$$

$$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$$

$$F_C = C\,\frac{dP_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{R}$$

Transformada de Laplace

$$F_a(s) = F_2(s) + F_L(s) = F_C(s) + F_R(s)$$

$$F_2(s) = \frac{P_a(s) - P_p(s)}{2}$$

$$F_L(s) = \frac{P_a(s) - P_p(s)}{Ls}$$

$$F_C(s) = C s P_p(s)$$

$$F_R(s) = \frac{P_p(s)}{R}$$

$$\frac{P_a(s) - P_p(s)}{2} + \frac{P_a(s) - P_p(s)}{Ls} = C s P_p(s) + \frac{P_p(s)}{R}$$

Procedimiento algebraico

$$\frac{P_a(s)}{2} - \frac{P_p(s)}{2} + \frac{P_a(s)}{Ls} - \frac{P_p(s)}{Ls} = C s P_p(s) + \frac{P_p(s)}{R}$$

$$\left(\frac{1}{2} + \frac{1}{Ls}\right) P_a(s) = \left(Cs + \frac{1}{R} + \frac{1}{2} + \frac{1}{Ls}\right) P_p(s)$$

$$\frac{Ls+2}{Ls} P_a(s) = \frac{CLRs^2 + L(2+R)s + R2}{LRs}\, P_p(s)$$

$$P_p(s) = \frac{\frac{Ls+2}{Ls}}{\frac{CLRs^2 + L(2+R)s + R2}{LRs}}$$

Resultado de la función de transferencia.

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + R2}{CLRs^2 + L(2+R)s + R2}$$

Error en estado estacionario

$$e(s) = \lim_{s \to 0} s P_a(s)\left(1 - \frac{P_p(s)}{P_a(s)}\right)$$

$$= \lim_{s \to 0} \frac{1}{s}\left(\frac{LRs + R2}{CLRs^2 + L(2+R)s + R2}\right)$$

$$= \emptyset$$

$$e(t) = \emptyset V$$

Estabilidad del sistema en lazo abierto.

Para analizar la estabilidad del sistema en lazo abierto.

$$\underbrace{CLR2s^2}_{a} + \underbrace{L(2+R)s}_{b} + \underbrace{R2}_{c} = \emptyset$$

$$\lambda_{1,2} = \frac{-L(2+R) \pm \sqrt{L^2(2+R)^2 - 4(CLR^2)2^2}}{2CLR2}$$

Hipotenso

$Z = 0.010$

$L = 0.001$

$R = 1.000$

$C = 3.500$

$$P_p(s) = \frac{LRs + R2}{CLR2s^2 + L(R+2)s + R2}$$

$$= \frac{0.001s + 0.01}{3.5\times10^{-5}s^2 + 0.00101s + 0.01}$$

$\lambda_1 = -14.429 + 8.8051i,$

$\lambda_2 = -14.429 + 8.8051i,$

Normotenso

$Z = 0.010$

$L = 0.001$

$R = 1.000$

$C = 1.500$

$$\frac{P_p(s)}{P_a(s)} = \frac{0.001 + 0.01}{1.5 \times 10^{-5} s^2 + 0.00101 + 0.01}$$

$\lambda_1 = -12.062$

$\lambda_2 = -55.272$

Estable y sobreamortiguada.

Hipertenso

$Z = 0.010$

$L = 0.001$

$R = 1.00$

$C = 1.000$

$$\frac{P_p(s)}{P_a(s)} = \frac{0.001 + 0.01}{0.00001 s^2 + 0.00101 s + 0.01}$$

$\lambda_1 = -10$

$\lambda_2 = -10$

Estable y criticamente amortiguada.

Modelo de ecuaciones integro-diferenciales

$$\frac{P_a(t) - P_p(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt = C\frac{dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

$$F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

La unica variable desconocida es la salida $P_p(t)$

$$\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - C\frac{dP_p(t)}{dt} = \frac{P_p(t)}{R} + \frac{P_p(t)}{Z}$$

$$P_p(t)\left(\frac{1}{R} + \frac{1}{Z}\right) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - C\frac{dP_p(t)}{dt}\right]$$

$$P_p(t)\frac{Z+R}{RZ} = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - C\frac{dP_p(t)}{dt}\right]$$

$$P_p(t) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - C\frac{dP_p(t)}{dt}\right]\frac{RZ}{Z+R}$$